

Dca7 (*S. pombe*, O74763) — Core-Function Hypothesis Review

Hypothesis: *Schizosaccharomyces pombe* Dca7 forms an evolutionarily conserved scaffold complex with a DYRK/Yak-family kinase. **Focus type:** core_function • **Gene:** dca7 / SPBC17D11.08 / UniProt O74763 (YBE8_SCHPO) • **Iteration:** 1 of 3

Executive Judgment

Verdict: Supported (strong for orthology/conservation; moderate for direct in-organism demonstration).

The hypothesis is well supported by two independent lines of evidence that converge:

1. **Orthology/evolutionary evidence (strong).** Dca7 is the fission-yeast ortholog of human **DCAF7/WDR68** (InterPro IPR045159 "DCAF7-like"; PomBase 1:1 orthologs = human DCAF7 HGNC:30915 and *S. cerevisiae* YPL247C). WDR68/DCAF7 is a textbook, deeply conserved seven-bladed WD40 β -propeller **scaffold for DYRK-family kinases** (DYRK1A/DYRK1B), documented in humans, zebrafish, insects and plants

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2. **Direct in-organism interaction (moderate).** In *S. pombe*, Dca7 physically binds **ppk15** (SPAC823.03), whose catalytic domain is specifically the **Yak1-type kinase domain** (CDD cd14212 PKc_YAK1) and whose orthologs are **human DYRK1A/DYRK1B and budding-yeast Yak1 (YJL141C)**. The Dca7–ppk15 interaction is a verified binary yeast-two-hybrid hit from the fission-yeast interactome (Vo et al., *Cell* 2016;

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Thus the two fission-yeast orthologs of the conserved human DCAF7–DYRK1A/1B pair physically associate in *S. pombe* — precisely the relationship the seed hypothesis predicts.

Three-species concordance (added Iteration 2). STRING confirms the DCAF7–Yak/DYRK partnership independently in two yeasts: *S. cerevisiae* **Yak1–YPL247C** combined score **0.996 (experimental subscore 0.979)** and *S. pombe* **dca7–ppk15** combined **0.933 (experimental subscore 0.873)**. Together with the direct mammalian DCAF7/WDR68–DYRK1A/1B binding (

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the kinase–scaffold pairing is experimentally supported across human, budding-yeast and fission-yeast lineages.

Sequence-conservation provenance (added Iteration 3). Global Needleman–Wunsch/BLOSUM62 alignments computed this run: **Dca7 vs human DCAF7 = 44.2% identity** (328 aligned cols), **Dca7 vs Sc YPL247C = 37.5%**, and **human DCAF7 vs YPL247C = 42.7%**. This ~38–44% identity across the shared ~330-residue WD40 β -propeller core, preserved over ~1 billion years, confirms Dca7 is a bona-fide DCAF7/WDR68 ortholog (not a name-only match). See [artifacts/conservation_provenance.md](#), [artifacts/evidence_matrix.csv](#), [artifacts/go_decision_table.csv](#).

GO annotation state (added Iteration 2). QuickGO for O74763 shows the molecular-function and biological-process roots are annotated ND ("**no data**," **GO_REF:0000015**) — Dca7 has **no experimental MF or BP annotation at all**. The only complex CC term, **GO:0080008 "Cul4-RING E3 ubiquitin ligase complex,"** is evidence code ISS, projected by similarity from human DCAF7 (UniProtKB:P61962). Nucleus (GO:0005634) is IBA (phylogenetic); cytoplasm/Golgi are IEA (UniProt-SubCell). So the DYRK/Yak-scaffold role is a genuine, unfilled annotation gap.

Most important caveats. (i) The pombe interaction rests on a **single high-throughput Y2H method**, without in-organism co-IP or a demonstrated scaffolding *function* (PomBase characterisation status = "conserved unknown"). (ii) "Scaffold" is an **inference from orthology**, not a measured pombe activity. (iii) The **Cul4-RING E3 complex CC annotation is an ISS carry-over from human DCAF7**, not pombe evidence, and is a plausible over-annotation relative to the better-supported DYRK-scaffold role.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	C
InterPro/UniProt O74763 (DB)	Computational (domain/ family)	Supports	Dca7 is a DCAF7/ WDR68-family WD40 protein	IPR045159 "DCAF7-like"; 7× WD40 (PF00400); β-propeller fold	S p
PomBase SPBC17D11.08 (DB)	Review/ database + orthology	Supports	Orthology to DCAF7/ WDR68	1:1 orthologs human DCAF7 (HGNC:30915), Sc YPL247C; "conserved in eukaryotes, single copy"	S
 26771498 (Vo et al., Cell 2016)	Interaction (binary Y2H)	Supports (direct, in-organism)	Dca7 binds a DYRK/Yak kinase in pombe	Verified binary Y2H: dca7 ↔ ppk15	S in
PomBase SPAC823.03 / CDD (DB)	Computational (domain/ orthology)	Supports	ppk15 is a Yak/DYRK kinase	Catalytic domain cd14212 PKc_YAK1 ; orthologs DYRK1A (HGNC:3091), DYRK1B (HGNC:3092), Yak1 (YJL141C)	S
 21777625 (Miyata & Nishida 2011)	Direct assay (co-IP, mapping)	Supports	DCAF7/ WDR68 is a conserved DYRK scaffold	WDR68 binds DYRK1A and DYRK1B (not DYRK2/3/4) via DYRK N-terminus; conserved WD40 protein	H m c

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	C
P 23349862 (Wang et al. 2013)	Mutant/ localization (in vivo)	Supports	WDR68 acts as a scaffold with Dyrk1	"highly conserved scaffolding protein... Ras-Map3k-Wdr68-Dyrk1 signaling relay"	Z C
P 25342745 (Miyata et al. 2014)	Direct assay (proteomics/ structure model)	Supports + qualifies	WDR68 binding repertoire & fold	Binds DYRK1A, MEKK1, and CUL4-DDB1 ; forms 7-bladed β -propeller	H c
STRING v12 (DB, aggregates experiments)	Interaction (multi-dataset)	Supports (independent)	DCAF7 ortholog binds Yak/ DYRK in yeast	Sc Yak1-YPL247C combined 0.996 (exp 0.979); Sp dca7-ppk15 combined 0.933 (exp 0.873)	S c &
This run (NW/ BLOSUM62; O74763,P61962,Q12523)	Structural/ evolutionary (computational)	Supports	Dca7 is a true DCAF7-family ortholog	Dca7-DCAF7 44.2% id (328 cols); Dca7- YPL247C 37.5%; DCAF7- YPL247C 42.7%	3 V
QuickGO O74763 (DB)	Review/ database (annotation state)	Qualifies (over- annotation flag)	Current GO annotations & evidence	MF & BP = ND ; GO:0080008 Cul4-RING = ISS from human DCAF7 (P61962); nucleus = IBA; cyto/Golgi = IEA	S
PomBase phenotypes (DB)	Mutant phenotype (HT)	Qualifies	Cellular role of dca7	Δ dca7: loss of viability in G0/stationary phase,	S

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	C
				decreased growth in glucose starvation; multiple stress-resistance phenotypes	

GO Curation Implications (leads — require curator verification)

- **Molecular Function (add — currently ND).** dca7's MF root is annotated ND ("no data"), so there is no existing MF to retain or conflict with. Evidence supports adding a **kinase-binding MF**. Recommended lead: **GO:0019901 "protein kinase binding"** (or the more specific **protein serine/threonine kinase binding**), evidence **IPI from**

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(dca7↔ppk15) supported by orthology (**ISO** to DCAF7–DYRK1A/1B). More informative than "protein binding" and should be preferred. No "DYRK binding" child term exists; "protein kinase binding" is the best-supported informative term.

- **Cellular Component (review; over-annotation confirmed as ISS carry-over).** QuickGO shows **GO:0080008 "Cul4-RING E3 ubiquitin ligase complex"** is evidence code **ISS**, projected by similarity from human DCAF7 (UniProtKB:P61962) — **not** pombe experimental data. In metazoa WDR68 does associate with CUL4-DDB1

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), so it is not wrong in principle, but for *S. pombe* it should be treated as **not experimentally established / non-core** and flagged for review. The **nucleus (GO:0005634)** term is **IBA** (phylogenetic), and UniProt independently annotates **cytoplasm + Golgi (IEA)**, so the subcellular picture is inference-only and partner-dependent; reconcile rather than assert nuclear localization. A **Dca7–ppk15 complex** CC could be a future lead but currently lacks a defined complex term.

- **Biological Process (candidate).** Consider a "**regulation of DYRK/Yak kinase signaling**"-type BP by orthology, but keep as **non-core/tentative** until pombe function is shown. $\Delta dca7$ phenotypes (G0/stationary-phase viability, glucose-starvation growth) are **downstream/pleiotropic** and should not be used to assign a direct MF/BP.
- **Net recommendation:** The seed hypothesis **should inform the review** by adding a kinase-binding MF lead and by scrutinizing the CUL4 CC annotation. Do **not** upgrade "scaffold" to an asserted pombe function without functional data.

Mechanistic Scope

- **Immediate molecular function being tested:** direct physical binding of the Dca7 WD40 β -propeller to a DYRK/Yak-family kinase (ppk15), i.e., a **kinase-scaffold/adaptor** activity.
- **Direct (supported):** Dca7 = DCAF7/WDR68 ortholog; Dca7 binds ppk15 (Yak/DYRK ortholog) in pombe; the human orthologous pair binds directly and the interaction is conserved.
- **Downstream / not the tested activity:** $\Delta dca7$ growth and stress phenotypes; "implicated in gene expression"; anthocyanin/craniofacial/developmental outcomes of orthologs; possible CUL4-DDB1 ubiquitin-ligase adaptor role. These are pathway consequences or context-specific roles, not the core scaffold activity.

Conflicts and Alternatives

- **Competing molecular role (CUL4 adaptor).** The "DCAF7 = DDB1-CUL4 associated factor" identity and the pombe CC annotation to a Cul4-RING E3 complex offer an alternative/additional function. Literature

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confirms WDR68–CUL4-DDB1 association in metazoa, so this is a genuine dual role, not simply an error — but which role is "core" in pombe is unresolved. The **DYRK-scaffold role is the more consistently documented, defining function** across the family.

- **Paralog/kinase-identity nuance.** *S. pombe* has more than one CMGC/DYRK-type kinase (e.g., **pom1**, MNB/DYRK subfamily; lkh1, LAMMER; prp4). The kinase captured interacting with Dca7 is specifically **ppk15 (Yak1/DYRK1A/1B ortholog)**, the expected DCAF7 partner

— this argues *against* paralog confusion, but a curator should confirm ppk15 (not pom1) is the relevant partner.

- **Localization conflict.** UniProt (cytoplasm + Golgi) vs PomBase GO CC (nucleus) — WDR68 orthologs are pan-cellular and shuttle with their partners

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so both may be partially correct in a partner-dependent manner.

- **Method limitation.** A single Y2H hit could in principle be a false positive; however, dca7 has only two recorded physical partners (not a sticky hub), which raises confidence.

Knowledge Gaps

1. **In-organism validation of Dca7–ppk15.** Checked: only one HT Y2H

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Matters because "scaffold" claims need biochemical confirmation. Resolve with **co-IP / affinity-MS of tagged Dca7 in pombe** and reciprocal ppk15 pulldown.

2. **Functional scaffolding readout.** Checked: PomBase status "conserved unknown"; no pombe assay. Matters for MF/BP assignment. Resolve by testing whether **Δdca7 alters ppk15 localization, stability, or substrate phosphorylation** (analogous to WDR68 controlling DYRK1A nuclear accumulation).
3. **Budding-yeast anchor.** *Largely resolved (Iteration 2):* STRING shows **Yak1–YPL247C** with combined 0.996 / experimental 0.979, strongly supporting the conserved pairing in *S. cerevisiae*. Remaining gap: the **specific primary reference(s)** behind that experimental subscore were not individually confirmed (likely large AP-MS/PCA datasets). Resolve via BioGRID/SGD YPL247C interaction records.
4. **CUL4 CC evidence code.** *Resolved (Iteration 2):* GO:0080008 is **ISS from human DCAF7 (P61962)** — a similarity carry-over, confirming the over-annotation concern. Curator action: decide whether to retain as non-core ISS or remove pending pombe evidence.

Discriminating Tests

- **Co-IP / proximity-labeling (BioID/TurboID) of Dca7 in *S. pombe*** to confirm ppk15 (and to distinguish ppk15 vs pom1 as the partner).
- **AlphaFold-Multimer / AF3 modeling of Dca7–ppk15** vs Dca7–DDB1 to compare interface plausibility (a fast in-silico discriminator between the DYRK-scaffold and CUL4-adaptor hypotheses).
- **Structure-guided mutagenesis** of the Dca7 β -propeller "top face" (the DYRK-binding surface in DCAF7) to test whether it is required for ppk15 binding and for Δ dca7 phenotypes.
- **Phospho-proteomics of Δ dca7 vs WT** to test whether Dca7 modulates ppk15/Yak-dependent phosphorylation (functional scaffolding).
- **Cross-complementation:** can human DCAF7 or the Dca7–ppk15 pair reconstitute known DYRK-scaffold outputs?

Curation Leads (require curator verification)

- **Candidate references to verify:**
 - PMID **26771498** (Vo et al., *Cell* 2016) — verify the exact Dca7 $\leftrightarrow$ ppk15 binary Y2H record in the supplementary interactome / BioGRID.
 - PMID **21777625** (Miyata & Nishida 2011) — snippet: "*DYRK1A and DYRK1B, but not DYRK2, DYRK3, or DYRK4, bound to endogenous and expressed WDR68.*"
 - PMID **23349862** (Wang et al. 2013) — snippet: "*Wdr68 is a highly conserved scaffolding protein... A Ras-Map3k-Wdr68-Dyrk1 signaling relay...*"
 - PMID **25342745** (Miyata et al. 2014) — snippet: "*WDR68 is an evolutionarily conserved WD40 repeat protein that binds to several proteins, including ... (DYRK1A), ... (MEKK1), and Cullin4-damage-specific DNA-binding protein 1 (CUL4-DDB1).*"
- **Candidate GO term changes:**
 - **Add MF:** GO:0019901 *protein kinase binding* (or protein serine/threonine kinase binding) — evidence IPI

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+ ISO (DCAF7). Preferred over "protein binding."

- **Review CC:** GO:0080008 *Cul4-RING E3 ubiquitin ligase complex* — flag as name/orthology-based; confirm evidence code; consider demoting to non-core for pombe. Reconcile nucleus (GO) vs cytoplasm/Golgi (UniProt).
 - **Optional BP (tentative, non-core):** regulation of DYRK/Yak kinase signaling — by orthology only.
 - **Suggested questions for curator:** Is ppk15 (not pom1) the intended DYRK/Yak partner? Is the Cul4 complex annotation experimentally grounded in pombe or an IEA carry-over? Should Dca7's core function be recorded as "DYRK-kinase scaffold/adaptor" pending functional data?
 - **Suggested experiments:** pombe co-IP/BioID of Dca7; AF-Multimer Dca7–ppk15; Δ dca7 phospho-proteomics; propeller-surface mutants.
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Provenance

All database results were retrieved programmatically this run: UniProt REST (O74763), PomBase API (SPBC17D11.08, SPAC823.03, SPAC25A8.01c), InterPro/CDD annotations, and NCBI eSummary for PMIDs. Interaction counts and orthology mappings are quoted directly from those queries. Human functional claims are from the cited primary papers (PMIDs above). No results were fabricated; where a resource was not queried (e.g., SGD YPL247C–Yak1), it is listed as a knowledge gap.
